Group 10 Homework 3

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1 Data collection, communication and storage

For this application, **MQTT** protocol is a better choice than **REST** for different reasons. The following points explain the reasons behind this, focusing on the tasks that this application carries out.

- measuring temperature and humidity every 2 seconds and publishing it into the Eclipse broker is a continous and repetitive task: this means that the **publish/subscribe** architecture of **MQTT** leads to a situation in which each subscriber receives the data as soon as it is published into the broker, without any other operation.
- the **Quality of service** supported by **MQTT** guarantees the delivery of the messages, which can be crucial for data collected by sensors where the loss or the duplication can be problematic; **REST** does not provide integrated mechanisms to guarantee the delivery of a message with the same reliability.
- in our specific case there are not multiple connections between the server and clients, but **MQTT** mantains an open attachment, reducing the latency (which is important since the measurements are collected and published every 2 seconds) and the overhead in the case of growth of the number of sensors and clients; the **REST** protocol, instead, requires the opening and the closing of a connection for each operation, raising the total latency.
- last but not least, the Topic mechanism provided by **MQTT** is a great choice to organize and deal with sensors, since the hierarchical nature of the topics make it easier to store data separetely for each sensor, quantity etc. and this makes it easier also to retrieve specific information by the client's side.

2 Data management and visualization

HTTP is used by the client to make different types of requests to the server. The most common requests are:

- **GET**: The GET HTTP method requests a representation of the specified resource. It is safe (i.e. doesn't alter the state of the server) and idempotent (i.e. the intended effect on the server of making a single request is the same as the effect of making several identical requests).
- **POST**: The POST HTTP method sends data to the server. It is not safe and idempotent.
- **PUT**: The PUT HTTP method requests that the data passed in the body payload be stored. It is not safe but it is idempotent.
- **DELETE**: The DELETE HTTP method deletes a specified resource. It is not safe but it is idempotent.

Given that, the most suitable HTTP method to retrieve temperature and humidity data is the GET method.